

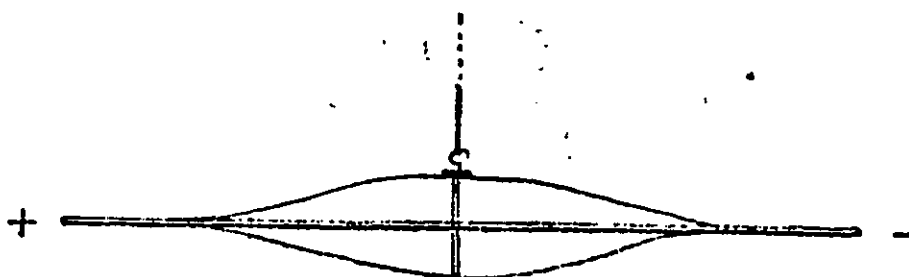
File: 24-185

Subject: SPECIAL INQUIRY

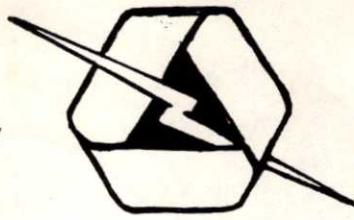
The Townsend Brown Electro-Gravity Device

A COMPREHENSIVE EVALUATION BY THE OFFICE OF
NAVAL RESEARCH, WITH ACCOMPANYING DOCUMENTS

15 SEPTEMBER 1952



STT. 17 86302



REX
RESEARCH

Box 1258

Berkeley CA 94701

INFOLIO

NR.

TITLE:

BROWN, Townsend

ELECTRO-GRAVITY

DEVICE REPORT

BERTRANDIAS/ CRAIG CONVERSATION (24 MAR 52): PAGE 2.

space with that instrument, it is a bad deal for somebody.

C- Well we will look into it Vic.

B- All right. The location of the place is, that I went to, is at (Beaumont?)- maybe Pat knows something about (it) but they down there told me that no-one else was in on it.

C- O.K. Vic. Good bye.

**
** CONFIDENTIAL SECURITY INFORMATION **
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Hq USAF

AFOSI-6

MAR 26 1952

PRIORITY

x

XXX

FROM: AFCSI-6

Request all information which can be developed concerning Townsend Brown Foundation, Los Angeles, California, through LAC and inquiry of Dunn and Bradstreet be forwarded this Headquarters by TWX not later than 27 March, 1952. No inquiries should be made beyond those requested above.

F.L. Welch
AFCSI-6

Code 192 x-176

Counter Intel. Div, OSI, IG

**
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TELEPHONE CONVERSATION BETWEEN MAJOR [REDACTED]

24 March 1952

General [REDACTED] - ***supporting a potential field something that I think I should mention to you. It sounds terribly screwy, but Friday I went down with Lehr, a man named Lehr, to a place called the Townsend Brown Foundation, and believe it or not, I saw a model of a flying saucer.

[REDACTED] - No.

B- Yes, this is going to sound very screwy to you, but I thought I should report it. Now I can't tell you much about the background, but this Townsend Brown Foundation has been working on this flying saucer development for a long while. There was a lot of objection to my getting in there by the party that took me and there were several people in there who identified themselves (censored) and some scientist, etc. and the thing frightened me- frightened me for the fact that it is being held or conducted by a private group. I was in there from about 1:30 until 5:00 in the afternoon and I saw these two models that fly and the thing has such a terrific impact that I thought we ought to find out something about it- who the people are and whether the thing is legitimate in development in this country or the formula is such that it is being conducted privately- if it ever gets away I say it is in the stage in which the atomic development was in the early days.

C- I see.

B- It was quite frightening. I made the inquiry whether the Air Force or the Navy knew anything about it and I was told- no. But I tell you after hearing it and all the other things that I had heard, I was quite concerned about it.

C- Well we will certainly take a look at it Vic.

B- I was told that I was not to say anything about it, but I'm afraid that all that I heard made me believe in it and these were not long-hand school boys- it was conducted in rather an elaborate office in Los Angeles and I understand that this man who had been at it for some 20 to 25 years and the way I evaluated it, if the Germans could have had a jet engine in the stage of war they did, whether or not someone could really have this further advanced than these people have. I thought I should report it to you.

C- Yes. Well I will look into it and see what I can find out.

B- Well it is the Townsend Brown Foundation, and there is a group of engineers- I don't know who these other people were at all. As I say, they were quite objective to my getting in there and seeing this demonstration.

C- And they are privately financed?

B- They are privately financed. I am of the opinion that if all I heard the other day- if it ever comes true, and somebody occupies

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AFCSI-6/Mr.Welch/ed/176/wrtn 28 Mar 52

DIRECTORATE OF SPECIAL INVESTIGATIONS
THE INSPECTOR GENERAL

AIR MAIL

AFCSI-6
24-185-18

31 Mar 1952

SUBJECT: TOWNSEND BROWN FOUNDATION, Los Angeles, Calif.
SPECIAL INQUIRY

TO:

[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

1. Reference is made to TWX, number 3-112, your office, dated 27 March 1952, in which you stated that a report, being compiled on the subject organization by Dunn and Bradstreet, would be completed in approximately two (2) weeks.

2. It is requested that you furnish this headquarters with a summary of the data being compiled by Dunn and Bradstreet just as soon as the information is available to your office. This headquarters is interested in receiving as much detail concerning the Foundation as can be acquired from Dunn and Bradstreet as a result of its compilation of data concerning the organization.

FOR THE DIRECTOR:

[REDACTED]
Chief, Counter Intelligence Div.
Dir. of Special Investigations
The Inspector General

AFCSI-6

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THE INSPECTOR GENERAL USAF
18th DIST. OFFICE OF SPECIAL INVEST.
AF Specialized Depot, P.O. Box 310
Maywood, Calif.

AIR MAIL

18D 24-169

SUBJECT: TOWNSEND BROWN FOUNDATION
Los Angeles, California
SPECIAL INQUIRY

3 April 1952

TO: Director of Special Investigations
Hq USAF, Washington 25,D.C.

1. In conformity with your letter of 31 March 1952, Headquarters file 24-185-18, attached is a copy of a Dun and Bradstreet credit report prepared on the above organization.

2. This District anticipates conducting no further inquiry on this matter.

1 Incl:

[REDACTED]
Col. USAF, District Cmdr.

Security Information - Confidential

DEPARTMENT OF THE AIR FORCE
STAFF MESSAGE DIVISION

INCOMING CLASSIFIED MESSAGE

Confidential
Security Information

Paraphrase not required. Notify
Cryptocenter before declassifying

From: 18 Dist. OSI, Maywood, Calif.

To: C/S, USAF, WASH, DC. ATTN: DIR OSI

Subject: Townsend, Brown Foundation 27 Mar 52 (DTL

Urmsg Townsend Brown Foundation LosA Calif. LAC negative.
Dunn and Bradstreet presently compiling credit rept on this orgn
which will be completed in Apr. 2 weeks. Advise if you desire
this office submit resume that report. No rept being written and
this case considered ruc to your Hqs.

ACTION: CSI
INFO: CDS, CIG

GRS/maj

(28 Mar 52)

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DIRECTORATE OF SPECIAL INVESTIGATIONS
THE INSPECTOR GENERAL
Washington 25, D.C.

AFCSI-6 24-185-18

28 March 1952

MEMORANDUM FOR GENERAL CARROLL:

SUBJECT: TOWNSEND BROWN FOUNDATION, Los Angeles, Calif.

1. Records of Headquarters FBI, ONI, G-2, and OSI do not reflect any information concerning the TOWNSEND BROWN FOUNDATION, which was the subject of the conversation between General Craig and General Bertrandias on 24 March 1952.

2. DO#18 has also advised that a local agency check was negative. However, it was determined that Dunn and Bradstreet at Los Angeles is presently compiling a credit report on this organization which will be completed in approximately two weeks. DO#18 is being requested to forward the substance of that report just as soon as it is available.

31 Mar
Gen. Jeter D: /A
advised of above. He requested
Dunn + Bradstreet report on
now as needed. AC.

Chief, Counter Intelligence Div.
Dir. of Special Investigations
The Inspector General

Confidential Security Information

AN INVESTIGATION RELATIVE TO THOMAS TOWNSEND BROWN

Office of Naval Research, Pasadena

1. INTRODUCTION

This report summarizes a technical investigation of certain claims of Thomas Townsend Brown concerning a proposed method of propulsion for aerial vehicles. The investigation, which began on May 3, 1952 and which is concluded hereby, is mainly analytical, no allotment having been made for an experimental program.

2. BRIEF DESCRIPTION OF THE ALLEGED PHENOMENA

Mr. Brown claims to have discovered several new physical effects, and considers them to be interrelated. We will be concerned here with those relating to the force upon and between electrically charged bodies in air or under oil.

2.1. It is asserted that the force exerted upon a negatively charged body by a positively charged body is greater than the force exerted upon the latter by the former; that the force differential is proportional to the mass of the dielectric inserted between the bodies; and that the phenomenon is evidence of a new "electrogravitic" effect. Paragraph 4 contains a fuller discussion of these phenomena.

2.2. A corollary effect is the force tending to move a metallic disc which by means of a plastic disc supports a fine wire, AA tangentially disposed in its plane, as shown in Fig. 1. When the wire is positively charged (e.g. 30 kilovolts) and the disc negatively charged, a propulsion force of a fraction of an ounce appears which moves the disc (and with it the wire) in the direction of the arrow C. These effects have been witnessed by the writer. Further discussion is presented below in paragraph 6.

2.3. An apparatus has been examined superficially, which shows an unexpected torque on an oil immersed disc when charged. This, according to Mr. Brown, is arranged to record automatically and frequently the magnitude of the angular deflection of the disc. These records have been tabulated and partially analyzed by Mr. Brown. Greater detail regarding this device is given in paragraph 5 below.

★ 2.4. There has been briefly inspected a third electrical device, purporting to be a communication equipment operating on waves related to the electrogravitic effect. It appears that the transmitter is simply a relaxation oscillator consisting of a high tension DC power supply, a high resistor, and a condenser which periodically discharges through a short gap. The receiver, shown in Fig. 2, consists of two Pyranol capacitors 0-5mf, two titanium oxide capacitors 500mmf, a Brush recorder, and a bypass capacitor 0.1 mf. to prevent the recorder from oscillating. Two terminals in the circuit are charged as indicated before reception and left floating during reception. The bridge of four

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FAX TRANSMISSION

625 Christian St Philadelphia, PA 19147 215/923-9341

TO:

..... AFMC/HO

SUBJECT:

..... TOWNSEND BROWN DOCUMENTS

..... Attached are relevant documents. I can't
..... vouch for authenticity, but they seem
..... plausible. Note the statement that as of
..... early 1952, "Winterhaven" was a self-funded
..... project. Since Brown was forced to go
..... to France in 1955 to continue his work,
..... I would assume that he tried to approach
..... DOD for funding but failed. In 1956,
..... of course, he helped found NICAP, the
..... UFO group. I hope this is helpful -
..... everything else I have is from commercial
..... magazines. Apologies for the delay.

..... - [REDACTED]

UNCLASSIFIED

24-185-18

(Unclassified) TOWNSEND BROWN FOUNDATION, Los Angeles, California
SPECIAL INQUIRY

Topical Intelligence Div., Deputy Dir. for Estimates
Directorate of Intelligence, DCS/O

Counterintelligence Div., Dir. of Special Investigations, IG

7 OCT 1952

1. Above subject was brought to the attention of this office by Major General V.E. Bertrandias following a visit to the Townsend Brown Foundation at Los Angeles, Calif. During this visit General Bertrandias saw a model of a "flying saucer" and engaged in conversation with Mr. Townsend Brown and other scientists who disclosed that they had been working on the project for twenty to twenty-five (20 to 25) years.

2. Attached herewith for your information are two (2) copies of a report by the Office of Naval Research relative to this matter and two (2) copies of a Dun & Bradstreet (sic.) Credit Rating pertaining to subject, dated 24 March 1952.

3. A check of local and national agencies was conducted with negative results. The files of this office contain no further pertinent data.

[REDACTED], Col. USAF
Acting Chief, Counterintelligence
Div., Directorate of Special
Investigations, Inspector Genl.

2 Incls.

AFCSI-6

Capt. Brown
Lt. Co. Free

UNCLASSIFIED

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CONFIDENTIAL SECURITY INFORMATION

24-185

(Confidential-Security Information) TOWNSEND BROWN FOUNDATION, Los Angeles, California - SPECIAL INQUIRY

Topical Intelligence Div. Deputy Dir. of Estimates
Directorate of Intelligence, DCS/O, ATTN: Maj Fournet

29 OCTOBER 1952

FROM: Counter Intel. Div., Directorate of Special Investigations, IG

1. Reference is made to R&R, this office, dated 7 October, 1952, concerning the above subject.

2. Forwarded herewith for your information is a Dun & Bradstreet (sic.) credit report dated 22 August 1952 which is a supplement to the Dun & Bradstreet credit rating forwarded to your office as an inclosure to the referenced R&R.

[REDACTED], Maj. USAF
Actg. Chief, Sabotage & Espionage
Branch.

1 incl. D & B Credit rtg. (quad)

AFCSI-6

Capt. Brown
Maj. Snipes

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DEPARTMENT OF THE AIR FORCE
HEADQUARTERS UNITED STATES AIR FORCE
WASHINGTON 25, D.C.

AFDRD

22 SEP 1952

MEMORANDUM FOR MAJOR [REDACTED]
Deputy Inspector General, USAF

SUBJECT: Townsend Brown Foundation

1. Reference is made to Memorandum from this office, same subject as above, dated 18 September 1952.

2. This office has been advised by Office of Naval Research letter, serial 23480, that the research report on the subject "An Investigation Relative to Thomas Townsend Brown" has been downgraded from CONFIDENTIAL to UNCLASSIFIED.

[REDACTED] JR.
Scientific Advisor
Directorate of Research &
Development
Office, Deputy Chief of Staff,
Development

UNCLASSIFIED

Lt. Col. Free/es/AFCSI-6
Code 192/x 140/6 Oct 52
24-185-18

CONFIDENTIAL SECURITY INFORMATION

(Unclassified) TOWNSEND BROWN FOUNDATION, Los Angeles, California
SPECIAL INQUIRY

The Office of the Inspector General, USAF
Norton Air Force Base, California

Counter Intelligence Div. Dir. of Special Investigations, IG

1. Reference is made to telephone conversation between Lt. General Craig and Maj. General Bertrandias on 24 Mar 52 concerning above subj

2. For your information there is attached a report by the Office of Naval Research, Pasadena, Calif. and a credit rating compiled by Dun & Bradstreet (sic.) in regard to this matter. Additionally, a check of local and national agencies concerning the Townsend Brown Foundation has been conducted with negative results.

3. The Directorate of Intelligence has been furnished all available information pertaining to this matter.

[REDACTED] Col. USAF
Acting Chief, CI Div.
Directorate of Special Invest

AFCSI-6
Lt.Col.Free

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6. THE MODEL FLYING SAUCERS

At the quarters of the Townsend Brown Foundation a room is devoted to demonstrating a new method of propulsion (paragraph 2.2.) which is claimed to depend upon the electrogravitic effect.

6.1. At the center of the room there stands a tall iron post carrying at its top a transverse, Plexiglass beam about six feet long, which is freely pivoted at its center to rotate in a horizontal plane. From each end of the beam hangs a model flying saucer, Fig. 1., of which the metal part is composed of two spun aluminum discs 12" in diameter, between which is bolted an 18" Plexiglass disc. The periphery of this disc is indented with nearly adjacent semi-circular scallops. The fingers remaining between the scallops constitute standoff insulators supporting a bare 0.005-inch stainless steel wire AA at a distance of 3 inches from the aluminum spinnings and insulate it from the aluminum. This will be called the outboard wire in the following discussions. It embraces perhaps one-quarter of the circumference (a similar wire, BB, is supported at the opposite quadrant, but is connected to the aluminum spinnings; it appears to be without effect). Each of these two identical assemblies is suspended by an insulating string from one end of the Plexiglass beam. By rubber-covered leads, shown in Fig.3., from the beam spinnings can be given one polarity while the outboard wires are given the opposite polarity. The leads are energized through rotating contacts at the pivot. If the beam rotates counter-clockwise as seen from above, the discs revolve about the post, in such a way that the outboard wire is at the leading edge.

A full-wave rectified unsmoothed high-tension supply in an adjacent room provides voltage for the demonstration. The voltage is controlled by a Variac. An examination showed no hidden connections.

6.2. When the power was turned on, the discs became negative and the outboard wires positive. Corona discharge could be heard as a gentle hiss or crackling. It could be seen in the dark, particularly around the outboard wire. As soon as the potential was applied, the discs and the beam began to revolve, the leading edge being at the outboard wires.

6.2.1. The thrust was roughly measured by a method in which the rotation was prevented, as shown in Fig.3. The weight on the pan was adjusted until the indicated angle was 45° , at which time the thrust was known to equal the force in the vertical thread. This method of thrust measurement is valid if the pivot is frictionless; actually the friction seems to introduce an uncertainty of about $\pm 0.5\%$ in the thrust data. The procedure described provides a value of the thrust of the two discs combined.

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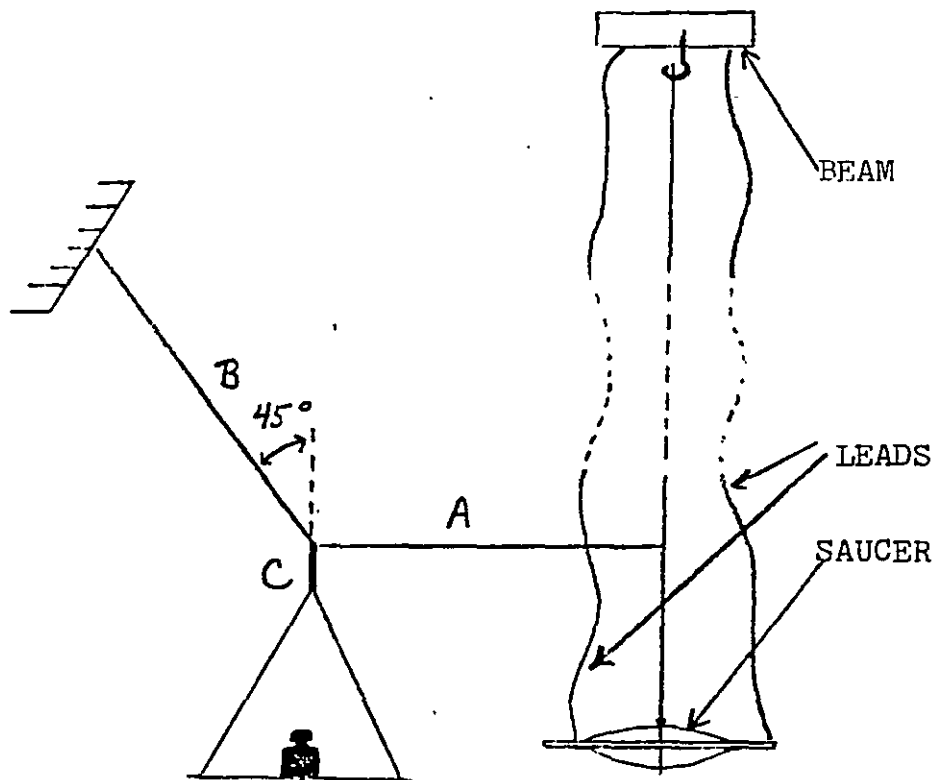


FIGURE 3. THE METHOD INVOLVING THREE THREADS A, B, AND C.
AS USED TO MEASURE THE THRUST OF A DEMONSTRATION
FLYING SAUCER

6.2.2. The data shown in Table 5 were gathered by the writer on 6 June with the collaboration of Mr. Brown and (censored), an assistant. Column 1 gives the setting of the Variac in the supply circuit for the high-tension transformers. The corresponding voltages, with respect to ground at the positive and negative outpost cables from the rectifier are given in columns 2 & 3; at each polarity the voltage was measured with the aid of a 0-2ma milliammeter and a high series resistor. The current in each lead recorded in Columns 4 & 5, was measured by the same millimeters when appropriately reconnected; it may be presumed that most of the current represented corona discharge between the discs and the outboard wires. Column 6 gives the measured thrust. In Column 7 is recorded the time taken for the discs to revolve once completely when free to do so, after they had had time to reach their full speed. The entires in Column 8 give the thrust to be expected on the basis that it is caused by the electric wind (see Par. 7 below). The last Column shows the efficiency of the propulsion, calculated (with due heed to the conversion of units) from the ratio of the electrical power input (Col. 2 x Col. 4 plus Col. 3 x Col. 5) to the mechanical power output (Col. 6 x Velocity). The data in Columns 2 & 3 were gathered first, then those in Columns 4, 5 & 6, and finally in Columns 2 & 3 again. The repeat measurements of the voltages nearly always agreed with the first measurements to within 0.75 kv or less. The average is recorded in Columns 2 and 3.

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<u>LUNAR HOUR</u>	<u>MEAN DEFL.</u>	<u>LUNAR HOUR</u>	<u>MEAN DEFL.</u>
0100	28.64	1300	28.59
0200	28.64	1400	28.62
0300	28.67	1500	28.65
0400	28.61	1600	28.61
0500	28.62	1700	28.62
0600	28.64	1800	28.62
0700	28.60	1900	28.62
0800	28.59	2000	28.62
0900	28.61	2100	28.66
1000	28.60	2200	28.64
1100	28.60	2300	28.60
1200	28.62	2400	28.63

TABLE 4. THE DEFLECTION OF THE INSTRUMENT AS AVERAGED OVER THE FIRST HALF OF 1937 FOR EACH LUNAR HOUR ANGLE

5.3. By the foregoing sketchy analysis of the data from the Brown Electrometer, the following conclusions are implied:

- (a) Through the weeks and months the data drift significantly;
- (b) There is no evidence of a solar diurnal period synchronizing with the sun, as claimed by Brown.
- (c) There is no evidence of a lunar diurnal period synchronizing with the moon, as claimed by Brown. On the contrary, the data appear most emphatically to deny such a period.

5.4. Two related experiments have been described by Mr. Brown which throw light on the reason for the deflections if not on the reason for their fluctuations.

5.4.1. In one, the sector disc was held stationary while a grounded metal disc was suspended horizontally over it in the oil, by a torsion fibre. When the sector disc was charged a motion of the adjacent oil was noted by Mr. Brown. The suspended disc deflected, but not when a glass plate was interposed between the sector disc and the suspended disc.

5.4.2. The other related experiment concerns an oil-immersed annulus described by Mr. Brown as 2 feet in diamtere and six inches thick, made of lead oxide(PbO) and parrafin in eight segments separated by eight copper strips alternately charged positive and negative. By means of a probe the potential distribution on the PbO and parrafin surfaces was explored and it was noted that whereas the gradient was uniform across the edges of the lead oxide sectors, the gradient on the parrafin was steepest near the negative electrodes. This observation means that the steepest gradients are all in the same direction, e.g. clockwise. It was also noted that the application of high voltage to the electrodes of this disc caused a lively motion of the oil.

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at lunar hour angle 1300 during the first half of 1937. This process was repeated for each of the 24 hours. The means are listed in Table 4. Their constance is most striking; they show no dependence upon lunar hour.

MONTH	HALF MONTH	RECORD FOR <u>0100</u>		RECORD FOR <u>1300</u>	
		Avg.Defl./Prob.Error		Avg. Defl./Prob.Error	
Jan	1st	26.05	✓ .15	26.25	✓ .21
	2nd	27.21	✓ .15	27.23	✓ .16
Feb	1st	28.43	✓ .22	28.17	✓ .09
	2nd	28.81	✓ .22	28.51	✓ .16
Mar	1st	29.79	✓ .15	29.98	✓ .13
	2nd	30.35	✓ .11	30.48	✓ .12
Apr	1st	30.25	✓ .10	29.96	✓ .10
	2nd	29.18	✓ .18	29.16	✓ .15
May	1st	28.43	✓ .14	28.24	✓ .18
	2nd	27.76	✓ .11	27.69	✓ .10
Jun	1st	27.81	✓ .17	27.85	✓ .09
	2nd	29.44	✓ .22	29.31	✓ .28

TABLE 2. HALF-MONTH MEAN DEFLECTIONS AT TWO TIMES A DAY, DURING THE FIRST HALF OF 1937

HOURL	AVG. DEFL.	HOURL	AVG. DEFL.
0100	28.23	1300	28.51
0200	28.29	1400	28.40
0300	28.32	1500	28.48
0400	28.33	1600	28.33
0500	28.38	1700	28.44
0600	28.32	1800	28.39
0700	28.37	1900	28.45
0800	28.34	2000	28.40
0900	28.43	2100	28.51
1000	28.51	2200	28.43
1100	28.64	2300	28.45
1200	28.62	2400	28.38

TABLE 3. HOURLY DEFLECTIONS AVERAGED OVER THE FIRST DAY OF THE HALF-MONTH IN JAN-JUNE 1937. THE PROBABLE ERRORS ARE ABOVE 0.46

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Aside from the reversal of sign, the deflection undergoes fluctuations whose character is less dramatic but still large, e.g. 10-50%. The cause of these fluctuations has been sought by Mr. Brown by seeking for correlations between them and various outside influences, such as magnetic storms, temperature, pressure, the time of the year, civil time, lunar time, sidereal time, and the Dow-Jones average. Only the last five correlations are claimed. Tabulated data are available only for the year 1937. Hourly recordings throughout that year are tabulated, for both civil and lunar time. The writer has seen no convincing evidence that the stock market and the automatic recorder are correlated, although it is claimed that a chi-square analysis has been performed, with positive results.

5.2. The undersigned has conducted a partial statistical analysis of the data from the automatic recorder for 1937 as follows:

5.2.1. Use was made of a table furnished by Mr. Brown entitled "Hourly Readings of Instrument, 1937" and particular attention was directed toward the readings of 0100, one hour after midnight, on each day of the first half of 1937. During the first half of January, for instance, the readings on days at this hour were 26.20, 25.90, 26.20, 26.10, 26.80, 25.60, 26.00, 25.40, 26.00, 26.00, 25.60, 26.60 and 25.90. The mean is 26.05, the standard deviation 0.55 and the probable error of the mean 0.15. Table 2 gives the half-monthly means and probably (sic.) errors for time 0100 and for time 1300. This table shows that there was a strong and real variation during the spring of 1937; but the great similarity between the readings at 0100 and 1300 implies that there is no conspicuous diurnal effect. (See also paragraph 5.2.2).

5.2.2. A direct search was undertaken by the writer to see whether the data for the first half of 1937 reveal any solar diurnal period. Attention was confined to the first day in each half-month. On these twelve selected days the readings at time 0200, for example, were 26.60, 26.00, 27.90, 29.00, 28.10, 29.80, 30.40, 29.80, 28.90, 27.50, 27.40 and 28.10. The mean was 28.29 and the probable error of the mean 0.46. The probable errors, as far as calculated, were all in this neighborhood. Table 3 gives the means calculated as in the above example. It is seen that there is no strong diurnal periodicity, with the dubious exception of a peak at 1100 hours; this result confirms that of paragraph 5.2.1.

5.2.3. The undersigned has scrutinized a table of data captioned "1937 Readings of Instrument with Lunar Hour Angle" in a search for evidence of a variation of the readings with the position of the moon, as rising, setting, etc. The form taken by this search was as follows. For the calendar month January the average deflection for each lunar hour was noted; e.g. at lunar hour angle 1300, when the moon was one hour past the meridian, the readings for January averaged 26.74. At the same lunar angle in the succeeding calendar months the average readings were 28.60, 30.18, 29.50, 28.02 and 28.52. The average of those six averages gives 28.59 as the mean of all observations taken

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a similar weight level with the first. By means of the wires the weights were electrified to a measured voltage; the corona current was not measured. An optical projection system provided on the wall an image of a scale and insulated pointers attached to the weights. The weights were either hollow (5 lb.) or solid (30 lb.). In certain of the observations the weights were linked by an insulating bar.

4.3. An examination of the data makes it clear that the inequality in deflections reported by Brown is still present here, though only to the extent of 10-30% (tests 2-4) and less than that under conditions where the corona discharge is reduced (of Tests 1 & 5).

When cigarette smoke was drifted into the vicinity of the wires it showed the existence of a wind. Just above the weights the wind blew from negative to positive, while higher on the wires the wind was opposite.

5. THE BROWN ELECTROMETER

In a small thermostated room in the second basement of the Banks-Huntley Building, (address censored), Los Angeles, there is a continuing experiment which shows an extension of the basic phenomenon. This apparatus is the Brown Electrometer. This device is housed in an oil-filled tank of half-inch steel, perhaps two feet in diameter and a foot deep, with a half-inch steel cover. In the tank there is suspended a horizontal disc 12" in diameter and one inch thick, made of six segments of pine wood alternating with six of marble around an 8" hub of Bakelite. The sections are separated by 1 x 2 x 1/16-inch copper strips. Alternate strips are grounded to a central metal stem supporting the disc which is attached to a music-wire suspension 0.018" in diameter and 6 3/8" long; the remaining copper strips connect to a central iron tit at the center of the hub, which dips into a cup of mercury. The disc is surrounded by a grounded metal shield which is attached to the tank and which stands at a clearance of 1/2-inch above and below the disc and 1 1/2-inch radially; this shield is centrally perforated at the top and bottom for the suspension and the tit.

A sequencing device in a cabinet below the tank causes a half-wave smoothed DC power supply to impress upon the mercury cup a well-regulated negative potential of 11.1 kv resulting in a leakage current of 0.45ma. The sequencer leaves the voltage on for 30 seconds, causing the disc to rotate several degrees. After 30 seconds, a short spark is made to jump from a point projecting about 9 inches horizontally in air from the stem of the disc, and perforates a rice-paper strip wound on a drum which rotates once each day. The voltage is then turned off and the disc returns to its relaxed position, which is recorded by a separate spark. The cycle repeats every three minutes. The apparatus appears to be well engineered and extremely reliable.

5.1. Mr. Brown has given attention for several years to the variations in the deflections of this electrometer. On the spark record the rest position is 50 units and the deflected reading is about 80. Until about a year and a half ago the deflected position was about 20, but suffered a gradual change to its present value at that time.

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The condenser dielectric was either homogenous or laminated. In one laminated version, 10,000 sheets of 0.001-inch lead foil were separated by an equal number of 3 x 3" sheets of 0.0008-inch cellulose acetate in a Formica box, to make 10,000 condensers in a series arrangement weighing 10kg. The acceleration exhibited by this device when suitably mounted is claimed to have been 2.5cm per sec² when the electrodes were charged, evidencing a force of 25,000 dynes or about 25 gm; the homogeneous type is stated by Mr. Brown to have had comparable thrust. The force was usually toward the positive electrode, although in some units it was toward the negative; this effect was attributed to a "polarization". The force is stated to have been stronger at some time than others, correlating with the position of the sun and moon. Data are displayed which relate to the angular speed of a pivoted gravitator, and which show half a dozen large fluctuations in a day; some of these can perhaps be construed as being dips simultaneous with the rise or setting of the sun or moon.

On 17 June 1952 the undersigned accompanied (censored) of the Townsend Brown Foundation to the Glendale plant of the Bendix Aviation Corporation. An experiment of the type described in paragraph 4.1 had been set up by Bendix personnel. (Sentence censored) The observations in Table 1 were recorded by the undersigned.

TEST NO.	WIRE LENGTH (in.)	WIRE SPACING (in.)	CYLINDER WEIGHT (lbs.)	VOLTAGE Pos.(kv)	VOLTAGE Neg.(kv)	MOTION Pos.(cm)	MOTION NEG.(cm)
1	66	8 1/2	5	30	30	.23	.23
2	66	6-1/8	5	30	30(less?)	.66	.65
3	66	6-1/8	5	30	30	.76	.99
4	66	6-1/8	5	30	30	.83	.93
5	66	6-1/8	5	15	15	.15	.16
6	66	6-5/8	5	(erratic) 30	30	.01 toward pos. (weights linked)	
7	66	6-5/8	5	DV=54		.04 toward pos. (weights linked)	
8	66	6-5/8	5	DV=54 (reversed)		.02 toward pos. (weights linked)	
9	60	5-5/8	30	10	40	.00	.2
10	60	5-5/8	30	0	42	.04	.06
11	60	5-5/8	30	0	42	.04	.06
12	66	6 1/2	30	0	42	.00	.00
						(weights linked)	

TABLE 1. OBSERVATIONS BY THE WRITER ON 17 JUNE 1952- EXPERIMENT AT BENDIX AVIATION CORP., GLENDALE, CALIFORNIA.

From a wooden scaffold a cylindrical brass weight 8" long and 4" in diameter was hung with axis vertical by means of about 5' of #14 Cu wire (0.06408" diameter). A similar wire about 6" away suspended a

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3. THE INVESTIGATIVE METHOD

This study is limited to the technical claims of Mr. Brown and is not concerned with matters of personality or personnel; or with the Townsend Brown Foundation. Although propulsion is the immediate center of interest, the claimed basic effect and its several manifestations are of the essence also. Therefore the analysis is not restricted to the propulsion apparatus.

Rough quantitative data have been taken by the writer on the force between charged bodies, and on the force on the disc shown in Fig.1. He has also carried out a brief examination of the recorded data on the oil immersed disc. Whenever possible, word-of-mouth data have been ignored. It is regrettable that during the time of the investigation the greater part of the experimental data gathered over the years by Mr. Brown are unavailable, being in storage elsewhere.

Full reliance is placed on the sincerity of Mr. Brown in his written reports of the data, but no reliance is placed on his interpretations. His orally reported data have been partially discounted as being less reliable than his written records. There is no evidence of conscious deceit.

Most of the interviews have taken place at the Townsend Brown Foundation, 306 North Vermont Avenue, Hollywood, California.

4. THE BASIC EFFECT

At Zanesville, Ohio in 1927, Mr. Brown at his then Laboratory of Research Physics, performed experiments on the force acting upon charged conductors. The apparatus took two forms, as described in a manuscript "Tapping Cosmic Energy", dated 28 October 1927, which was submitted at that time to the Physical Review for publication but which was rejected. Apparently no basic experiments have been performed since 1928.

4.1. In one form the conductors were two independent spheres suspended close but not touching by means of fine wires which also served as electrical leads. The spheres were of lead, and weighed 14,740gm each. The wires supporting them hung from the ceiling 45cm apart. When the 127" spheres were oppositely charged to a potential difference of 125kv they moved toward each other slightly in response to their electrostatic attraction. The negative sphere was displaced allegedly about twice as far as the positive sphere. The inequality of the displacements is the essence of the phenomenon.

4.2. In the other form of the basic experiments the condenser was formed by a block of insulating material $3\frac{1}{2} \times 2\frac{1}{2} \times 15$ ", to the ends of which electrodes were attached. Whether the condenser was suspended or whether it was mounted at the end of a centrally pivoted horizontal beam, it showed a force causing the arm to rotate, usually in such a direction that the positive electrode moved forward. The force is claimed to have been proportional to the mass of the block, although no data are given in the manuscript and none are available. The name "gravitator" is used for this apparatus.

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capacitors is housed in a cabinet, indicated by a dashed line, which forms an obviously imperfect shield against electromagnetic radiation. when the transmitter is operated in an adjoining room, the Brush recorder jumps with each spark. Mr. Brown states that the effect is present, but reduced, if the 25kv potentials are not applied. If the Pyranols are short circuited and removed, the effect is still reported present. No attempt has been made at conscioncious electromagnetic shieldings of either the transmitter or the receiver. In the writer's opinion the response may be the result of a chance rectifying contact. The relation between the communication device and the mechanical effects which form the topic of this report appears so remote that no further mention will be made of the former.

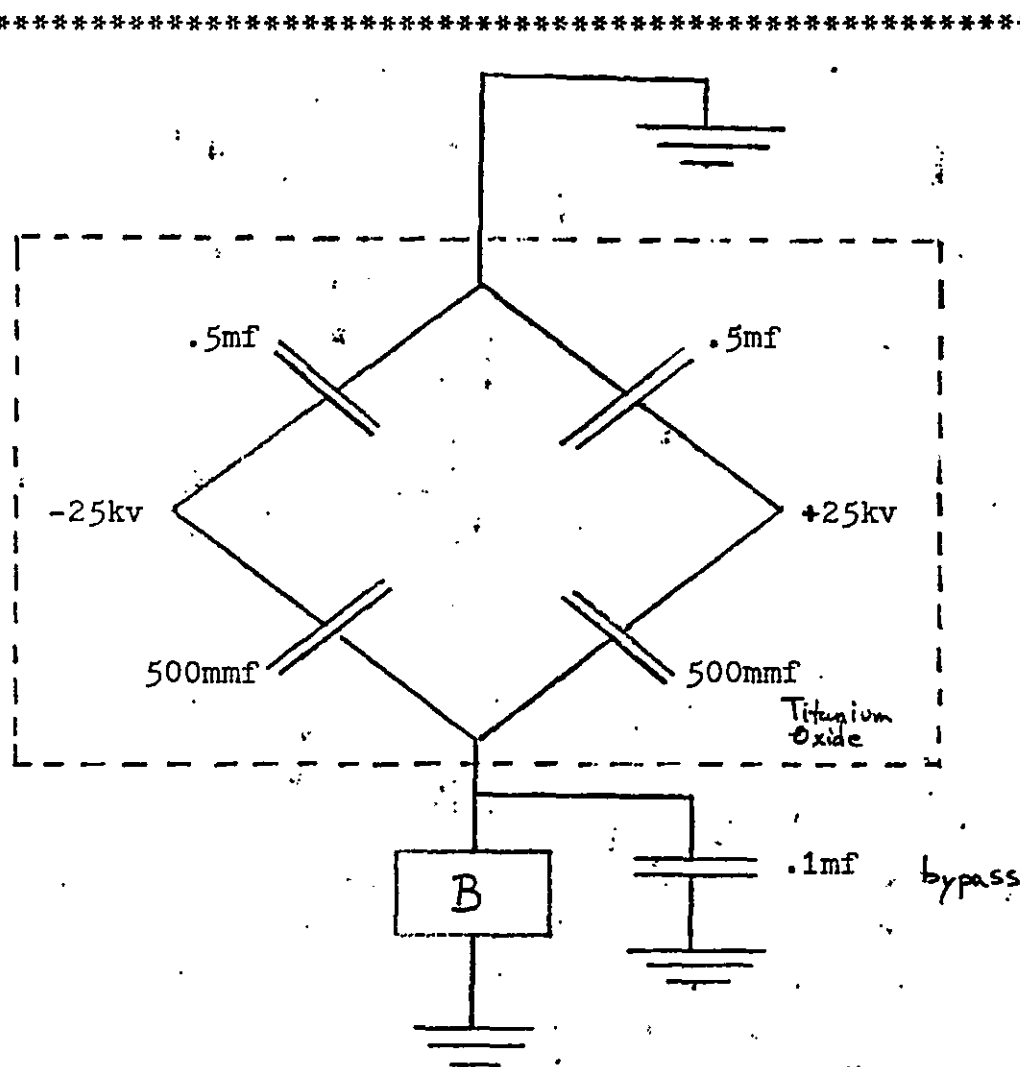


Fig. 2. The electrogravitic communication receiver

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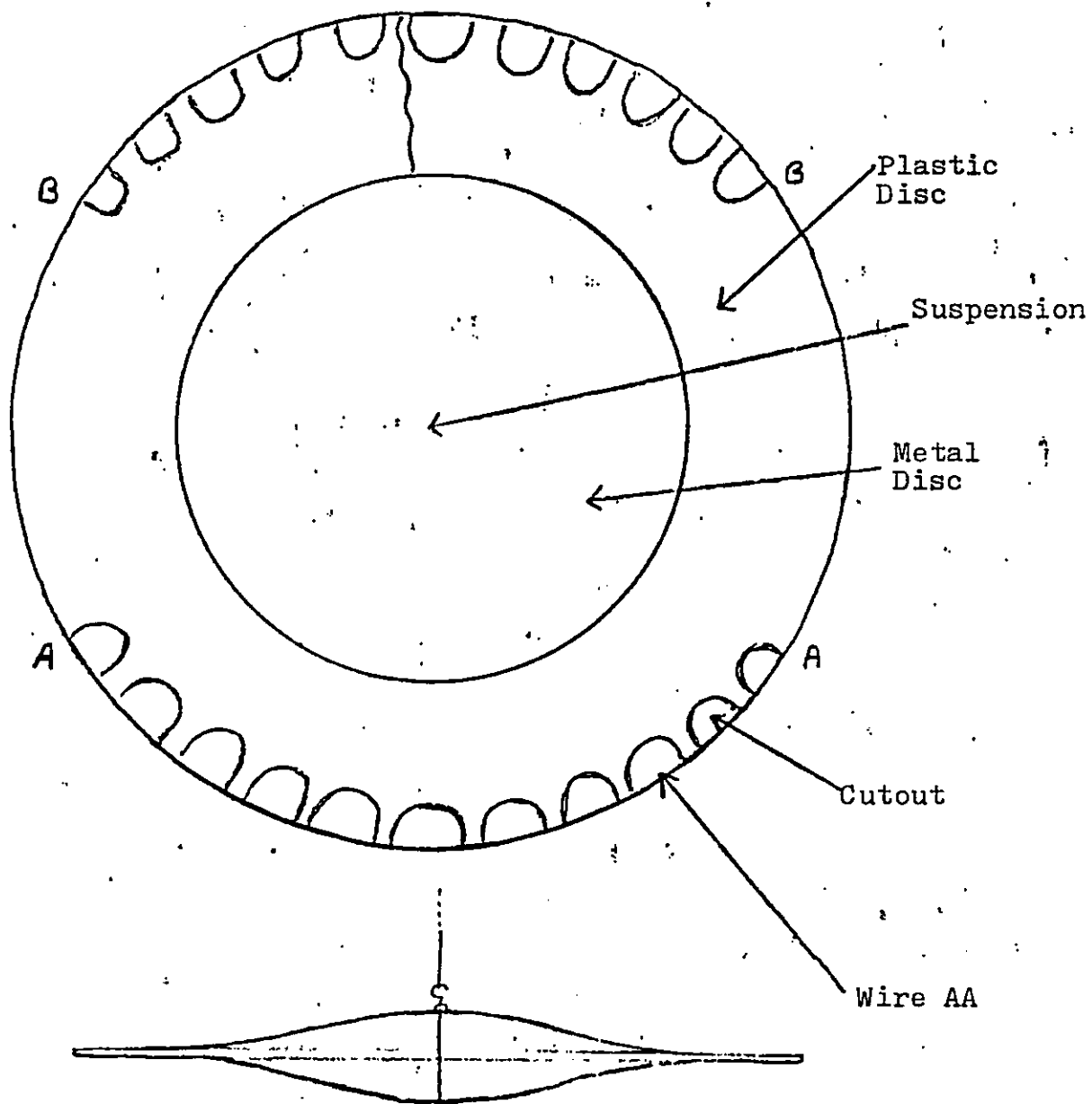


Fig.1. Demonstration Flying Saucer